Difference Equation and Filter Implementation

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Difference Equation

- **Any discrete-domain LTI/LSI system can be implemented using a difference equation that describes the input and output relationship:**

$$y[n] = \underbrace{\sum_{l} a_l x[n-l]}_{\text{feed forward}} + \underbrace{\sum_{m \neq 0} b_m y[n-m]}_{\text{feedback}}$$

- **When $b_m = 0$: FIR system, $a_l = h_l$ (impulse response).**

- **When $b_m \neq 0$: IIR system.**

- **When $a_l = 0$ except for a_0: pure IIR system, no feed forward:**

$$y[n] = a_0 x[n] + \sum_{m \neq 0} b_m y[n-m]$$

Question

- **For a discrete-time system described by the difference equation**

$$y[n] = 2x[n-2] + x[n-1] + y[n-1] - 3y[n-2]$$

- **Is it a FIR filter or an IIR filter?**

a) IIR

b) FIR

Example: Echo

- $y[n] = x[n] + ay[n - D]$
  - *a*: attenuation coefficient
  - *D*: the number of delayed samples, chosen depending on the echo depth required.

- **What type of system, FIR or IIR?**

- **Echo effect is generated by adding to the current input a delayed and attenuated version of a past output.**

- **How to implement?**

Echo: Implementation Using Circular Buffer

- $y[n] = x[n] + ay[n-D], D = 4, a = 0.5, x[n] = [1,2,3,4,1,2,3,4], n = 0, \dots, 7$
  - Implementation: $y[n\%D] = x[n] + ay[(n-D)\%D]$

Y[0]	Y[1]	Y[2]	Y[3]
0	0	0	0
1	0	0	0
1	2	0	0

Exercise: Fill in the rest of the table.

y buffer (size *N* = 4) is initialized with zeros.

$y[0\%D] = x[0] + ay[(-4)\%D] \Rightarrow y[0] = x[0] + ay[0] = 1$

$y[1\%D] = x[1] + ay[(-3)\%D] \Rightarrow y[1] = x[1] + ay[1] = 2$

Echo: Implementation Using Circular Buffer

- $y[n] = x[n] + ay[n-D], D = 4, a = 0.5, x[n] = [1,2,3,4,1,2,3,4], n = 0, \dots, 7$
  - Implementation: $y[n\%D] = x[n] + ay[(n-D)\%D]$

Y[0]	Y[1]	Y[2]	Y[3]
0	0	0	0
1	0	0	0
1	2	0	0
1	2	3	0
1	2	3	4
1.5	2	3	4
1.5	3	3	4
1.5	3	4.5	4
1.5	3	4.5	6

Answer:

y buffer (size *N* = 4) is initialized with zeros.

$y[0\%D] = x[0] + ay[(-4)\%D] \Rightarrow y[0] = x[0] + ay[0] = 1$

$y[1\%D] = x[1] + ay[(-3)\%D] \Rightarrow y[1] = x[1] + ay[1] = 2$

$y[2\%D] = x[2] + ay[(-2)\%D] \Rightarrow y[2] = x[2] + ay[2] = 3$

$y[3\%D] = x[3] + ay[(-1)\%D] \Rightarrow y[3] = x[3] + ay[3] = 4$

$y[4\%D] = x[4] + ay[0\%D] \Rightarrow y[0] = x[4] + ay[0] = 1.5$

$y[5\%D] = x[5] + ay[1\%D] \Rightarrow y[1] = x[5] + ay[1] = 3$

$y[6\%D] = x[6] + ay[2\%D] \Rightarrow y[2] = x[6] + ay[2] = 4.5$

$y[7\%D] = x[7] + ay[3\%D] \Rightarrow y[3] = x[7] + ay[3] = 6$

Echo: Implementation Using Circular Buffer

- $y[n] = x[n] + ay[n - D]$
- **Suppose the sampling frequency is 8192 Hz, for a 0.5 second delay, what is the value of *D*?**
- **How to choose the buffer size *N* to implement the echo?**

Example: Filter

- $y[n] = \sum_{l=0}^{L-1} a_l x[n-l]$
  $= a_0 x[n] + a_1 x[n-1] + \cdots + a_{L-1} x[n-L+1]$
- $a_l, l = 0, \ldots, \mathrm{L}-1$: **filter coefficients**
- **What type of system, FIR or IIR?**
- **How to implement?**

FIR Filter: Implementation

$$y[n] = a_0 x[n] + a_1 x[n-1] + \cdots + a_{L-1} x[n-L+1]$$

Filter Coefficients	a[0]	a[1]	a[2]	…	a[L-2]	a[L-1]
Current Time, n	x[n-L+1]	x[n-L+2]	…	x[n-2]	x[n-1]	x[n]
Next Time, n+1	x[n-L+2]	x[n-L+3]	…	x[n-1]	x[n]	x[n+1]

Discarded

New Data

FIR Filter: Implementation Using Circular Buffer

- $y[n] = a_0 x[n] + a_1 x[n-1] + \cdots + a_{L-1} x[n-L+1]$
- **More efficient implementation: move pointer, instead of move data.**

Filter Coefficients	a[0]	a[1]	a[2]	…	a[L-2]	a[L-1]
Current Time, n	x[n-L+1]	x[n-L+2]	…	x[n-2]	x[n-1]	x[n]
Next Time, n+1	x[n+1]	x[n-L+2]	…	x[n-2]	x[n-1]	x[n]

New data replace oldest data

FIR Filter: Implementation Using Circular Buffer

$y[n] = x[n] + 0.75x[n-1] + 0.5x[n-2] + 0.25x[n-3], x[n] = [1,2,3,4,4,3,2,1], n = 0, \ldots, 7$

- Store *x[n]* at pointer location *P.*
- Calculate output: $y[n] = x[P] + 0.75x[(P-1)\%N] + 0.5x[(P-2)\%N] + 0.25x[(P-3)\%N]$
- Increment pointer: *P = (P+1)%N*

	X[0]	X[1]	X[2]	X[3]
	0	0	0	0
P →	1	0	0	0
	1	→ 2	0	0

Exercise: Fill in the rest of the table and compute *y*.

x buffer (size *N* = 4) is initialized with zeros. *P* = 0.

$y[0] = 1$

$y[1] = 2 + 0.75 * 1 = 2.75$

FIR Filter: Implementation Using Circular Buffer

$y[n] = x[n] + 0.75x[n-1] + 0.5x[n-2] + 0.25x[n-3], x[n] = [1,2,3,4,4,3,2,1], n = 0, \ldots, 7$

- Store *x[n]* at pointer location *P.*
- Calculate output: $y[n] = x[P] + 0.75x[(P-1)\%N] + 0.5x[(P-2)\%N] + 0.25x[(P-3)\%N]$
- Increment pointer: *P = (P+1)%N*

	X[0]	X[1]	X[2]	X[3]
	0	0	0	0
P →	1	0	0	0
	1	→ 2	0	0
	1	2	→ 3	0
	1	2	3	→ 4
→	4	2	3	4
	4	→ 3	3	4
	4	3	→ 2	4
	4	3	2	→ 1

Answer:

x buffer (size *N* = 4) is initialized with zeros. *P* = 0.

$y[0] = 1$

$y[1] = 2 + 0.75 * 1 = 2.75$

$y[2] = 3 + 0.75 * 2 + 0.5 * 1 = 5$

$y[3] = 4 + 0.75 * 3 + 0.5 * 2 + 0.25 * 1 = 7.5$

$y[4] = 4 + 0.75 * 4 + 0.5 * 3 + 0.25 * 2 = 9$

$y[5] = 3 + 0.75 * 4 + 0.5 * 4 + 0.25 * 3 = 8.75$

$y[6] = 2 + 0.75 * 3 + 0.5 * 4 + 0.25 * 4 = 7.25$

$y[7] = 1 + 0.75 * 2 + 0.5 * 3 + 0.25 * 4 = 5$

FIR Filter: Implementation Using Circular Buffer

- $y[n] = a_0 x[n] + a_1 x[n-1] + \cdots + a_{L-1} x[n-L+1]$
- **How to choose the buffer size *N* to implement the filter?**

How to Get Filter Coefficients?

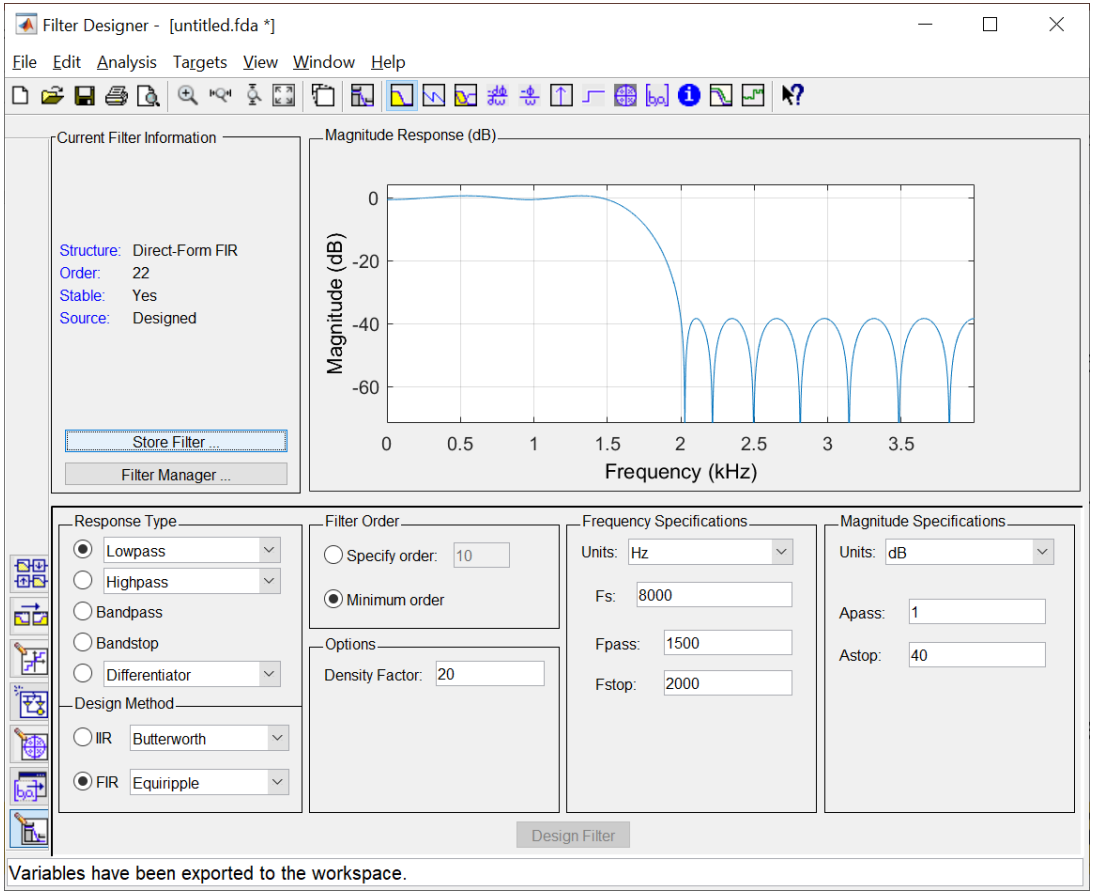
MATLAB

- Apps -> Signal Processing and Communications -> Filter Designer
- Or at command line type: filterDesigner

Example: design a FIR low pass filter with the following parameters

- Sampling frequency: 8000 Hz
- Passband frequency: 1500 Hz
- Stopband frequency: 2000 Hz
- Stopband attenuation: 40 dB

How to Get Filter Coefficients?



File -> Export

Num =

Columns 1 through 12

0.0132 0.0044 -0.0225 -0.0364
-0.0047 0.0393 0.0208 -0.0592
-0.0791 0.0681 0.3079 0.4269

Columns 13 through 23

0.3079 0.0681 -0.0791 -0.0592
0.0208 0.0393 -0.0047 -0.0364
-0.0225 0.0044 0.0132

$= [a_0, a_1, a_2, \ldots, a_{22}]$

What is buffer size *N*?

Online FIR Filter Design Tool

- **http://t-filter.engineerjs.com/**
- **Filter coefficients**
  - have both double and Q15 formats.
  - Can be easily copied and pasted.

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